

Aim

To develop a Student Performance Prediction System using the Multiple Linear Regression algorithm and evaluate its performance using the R^2 Score.

Software Requirements

- Python 3.x
- Visual Studio Code / Jupyter Notebook
- Pandas
- Scikit-learn
- UCI ML Repository ([ucimlrepo](#))

Dataset Used

Dataset Name: Student Performance Dataset

Source: UCI Machine Learning Repository

Dataset ID: 320

The Student Performance dataset contains multiple academic, demographic, and personal attributes of students. These features are used to predict the students' final academic performance using the Multiple Linear Regression algorithm.

Python Code

```
import pandas as pd
from ucimlrepo import fetch_ucirepo
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import r2_score

student_performance = fetch_ucirepo(id=320)

X = student_performance.data.features
y = student_performance.data.targets

X = pd.get_dummies(X, drop_first=True)

if isinstance(y, pd.DataFrame):
    y = y.iloc[:, 0]

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

model = LinearRegression()
model.fit(X_train, y_train)
```

```

y_pred = model.predict(X_test)

print("R2 Score:", r2_score(y_test, y_pred))

result = pd.DataFrame({
    "Actual": y_test.values,
    "Predicted": y_pred
})

print(result.head(10))

```

Output

- Dataset Loaded Successfully
- R² Score
- Actual vs Predicted Values

R ² Score: 0.1366313828682093		
	Actual	Predicted
0	17	11.844152
1	11	12.039095
2	18	12.612575
3	10	11.212347
4	13	10.275632
5	16	11.989627
6	17	13.201498
7	10	7.205994
8	10	11.344648
9	11	11.828469